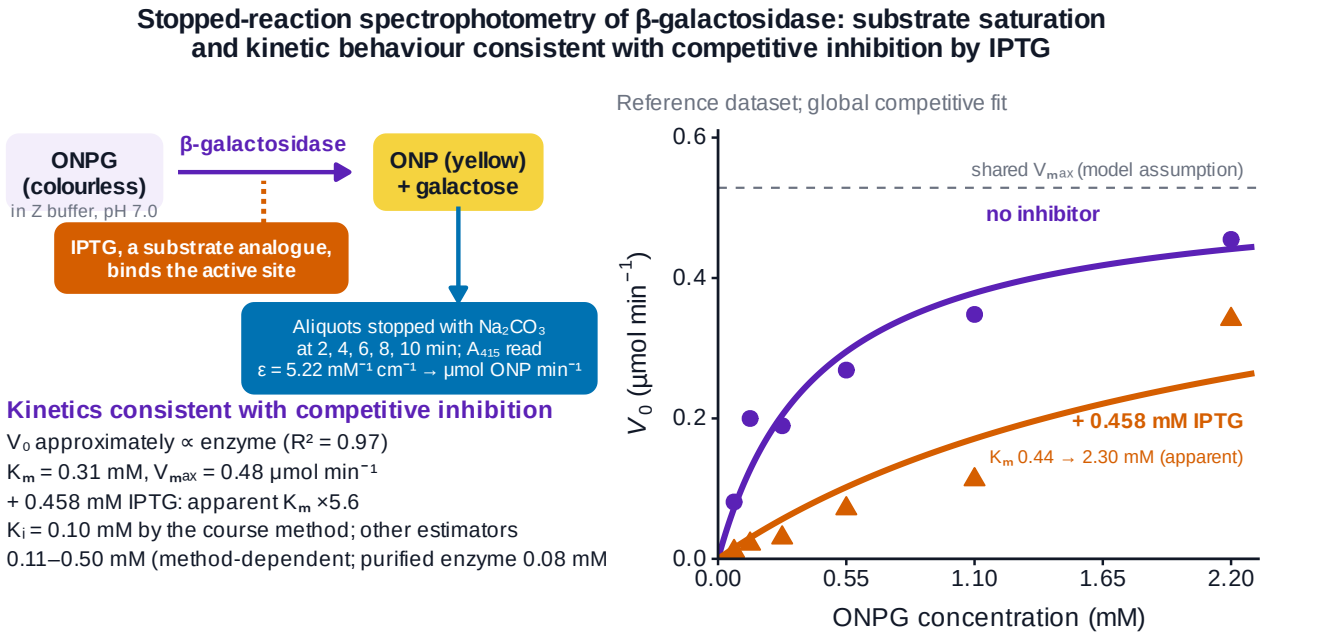
Article

Stopped-reaction kinetics of *E. coli* β-galactosidase with ONPG: substrate saturation and inhibition by IPTG consistent with a competitive mechanism

Graphical abstract



Highlights

- ONP calibration at 415 nm (ε = 5.22 mM^{-1} cm^{-1}) converts absorbance rates to product formed
- β-Galactosidase velocity rises approximately with enzyme amount and saturates with ONPG (K_m ≈ 0.3–0.6 mM)
- IPTG raises the apparent K_m 5.6-fold; a competitive model fits both series better than a noncompetitive one
- K_i = 0.10 mM by the course method; alternative estimators give 0.11–0.50 mM (method-dependent estimates, not a confidence interval), the same order as the purified-enzyme value

In brief

Using a stopped-reaction assay with the chromogenic substrate ONPG and a course-provided reference dataset, the authors measure the Michaelis constant and limiting velocity of *E. coli* β-galactosidase and show that the *lac* inducer IPTG raises the apparent K_m and inhibits most where substrate is lowest, kinetic behaviour consistent with competitive inhibition, with a K_i of the same order as that of the purified enzyme. A single explicit rule for choosing initial-rate windows makes every velocity reproducible from the raw absorbances, and a sensitivity analysis shows which conclusions survive alternative rules and which, including the numerical K_i, do not.

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Article

Stopped-reaction kinetics of *E. coli* β-galactosidase with ONPG: substrate saturation and inhibition by IPTG consistent with a competitive mechanism

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Code and data: see data and code availability

SUMMARY

Enzyme kinetics condenses how an enzyme responds to substrate, inhibitors and its own abundance into two constants, K_m and V_{max}. Here we characterize *Escherichia coli* β-galactosidase acting on *o*-nitrophenyl-β-D-galactopyranoside (ONPG) by stopped-reaction spectrophotometry, following the product *o*-nitrophenol at 415 nm after quenching timed aliquots with Na_2CO_3. The primary kinetic analysis uses a course-provided reference dataset; measurements collected by our own group form an experimental dataset analyzed separately. A Beer–Lambert calibration (ε = 5.22 mM^{-1} cm^{-1}) converts absorbance rates into product formed. Initial velocity rose approximately in proportion to enzyme concentration, and velocity saturated with ONPG (K_m = 0.31–0.36 mM, V_{max} = 0.48–0.49 µmol min^{-1} per assay). The gratuitous *lac* inducer isopropyl-β-D-thiogalactopyranoside (IPTG) raised the apparent K_m 5.6-fold and inhibited most where substrate was lowest; a competitive model fitted both series with a residual sum of squares 1.9 times smaller than a pure noncompetitive model, behaviour consistent with competitive inhibition. K_i depends on the model and on how initial rates are estimated from 2-min samples (0.10–0.22 mM across rule-based estimators), the same order as the 0.08 mM reported for the purified enzyme. An explicit rule for initial-rate windows makes every velocity reproducible from the raw absorbances.

INTRODUCTION

Enzymes accelerate reactions by binding substrates at active sites, and the dependence of velocity on substrate concentration reveals how tightly and how productively they do so. Michaelis and Menten described this dependence with a reversible enzyme–substrate complex that breaks down to product, giving $V_0 = V_{max}[S]/(K_m + [S])$.[1,2] $V_{max} = k_{cat}[E]_T$ is the limiting velocity when every active site is occupied, and is therefore proportional to the amount of enzyme; K_m is the substrate concentration at half-maximal velocity and reports how much substrate is needed to engage the enzyme. Lineweaver and Burk rearranged the equation into a straight line, $1/V_0 = (K_m/V_{max})(1/[S]) + 1/V_{max}$, whose intercepts give the two constants.[3] The reciprocal transform distorts the error structure of the data,[4] but it remains the standard teaching representation; we report it alongside a direct fit of the untransformed data.

β-Galactosidase (EC 3.2.1.23), the *lacZ* product of *Escherichia coli*, is a homotetramer of 1023-residue subunits that hydrolyzes the β-1,4 glycosidic bond of lactose and of synthetic β-galactosides.[5,6] The chromogenic substrate ONPG is cleaved to galactose and *o*-nitrophenol. At alkaline pH, *o*-nitrophenol exists as the yellow *o*-nitrophenolate anion, whose absorbance near 415–420 nm follows the Beer–Lambert law. Adding 1 M Na_2CO_3 to a timed aliquot therefore both stops the reaction, by raising the pH far above the enzyme's working range, and develops the color, so one reading gives the product formed up to that moment.[7] A series of stopped samples builds a progress curve whose initial slope is the initial velocity V_0, measured before substrate depletion, product accumulation, or enzyme instability slow the reaction.

Inhibitors reveal themselves by how they change K_m and V_{max}. A competitive inhibitor binds free enzyme at the active site and is displaced by substrate: it leaves V_{max} unchanged but raises the apparent Michaelis constant to $K_{m,app} = K_m(1 + [I]/K_i)$, where K_i is the inhibitor's dissociation constant. A noncompetitive inhibitor inactivates a fraction of the enzyme whatever the substrate concentration, lowering V_{max} and leaving K_m unchanged.[2] IPTG is the thiogalactoside used as a gratuitous inducer of the *lac* operon because it induces the enzyme without being hydrolyzed by it.[8,9] The same stability makes it a dead-end ligand of β-galactosidase: it occupies the galactose subsite of the active site, but its thioglycosidic bond is not cleaved, so it is expected to behave as a competitive inhibitor with a K_i of the order of 0.1 mM.[6,10]

Here we establish the ONP absorption spectrum and standard curve, measure the dependence of V_0 on enzyme concentration at saturating ONPG, and measure V_0 over a 32-fold range of ONPG concentrations with and without IPTG. Because five of the six uninhibited progress curves had already begun to decelerate within the first sampling interval, we formalized the choice of initial-rate window as an explicit, reproducible rule, tested its origin assumption, and quantified how much each kinetic constant depends on it.

RESULTS

ONP absorbs maximally at 415 nm and calibrates absorbance to product

The spectrum of 0.333 mM ONP in the alkaline assay matrix peaked at 415 nm (A = 1.723), with 410–420 nm within 2% of the maximum (Figure 1A), so 415 nm was used for all kinetic measurements. The course's standard-curve table listed seven readings against tubes 1–7 while the manual describes a blank and seven standards; the first reading (0) is the zero-concentration preparation, as the instructor confirmed, and the remaining six were assigned to the manual's standards 1–6, an assignment anchored by the spectrum tube (manual standard 4, which read 1.723 at 415 nm, identical to the

fifth reading; Table S1). The manual's standard 7 (0.667 mM) has no reported reading and no value was assumed for it. Standards of 0.083–0.433 mM ONP gave a linear Beer–Lambert plot, A_{415} = 5.222[ONP] − 0.022 (R^2 = 0.9988, n = 5); the 0.533 mM standard (A = 3.24) lay 17% above the line and was excluded (Figure 1B). The slope gives ε = 5.22 ± 0.10 mM^{-1} cm^{-1} (5222 M^{-1} cm^{-1}; 5.15 through the origin); including the excluded standard would give 6.01 mM^{-1} cm^{-1} and lower every V_{max} by 13% without affecting K_m or K_i. The excluded reading transmits 0.06% of the incident light, within but near the instrument's specified linear range (up to 3.5 A at 260 nm);[13] because the deviation is upward, the opposite of stray-light saturation, its cause cannot be identified from a single reading. Every velocity below was converted with ε: a rate of 0.132 ΔA_{415} min^{-1} (tube A, no inhibitor) corresponds to 0.132/5.22 = 0.0253 mM min^{-1} in the cuvette, × 1.5 (stop dilution) = 0.0379 mM min^{-1} in the reaction, × 12.0 mL = 0.455 µmol ONP min^{-1} per assay.

Initial velocity scales approximately with enzyme concentration

With 2.20 mM ONPG, product formation was linear over an initial window at every enzyme level (Figure 2A). V_0 fell from 0.469 µmol min^{-1} (0.136 ΔA_{415} min^{-1}) with the undiluted enzyme to 0.079 µmol min^{-1} at 0.15 of that amount, a 5.9-fold change for a 6.7-fold dilution; relative rates were 1.00, 0.77, 0.60, 0.18, 0.17 for relative enzyme 1.00, 0.75, 0.50, 0.25, and 0.15. A line through the origin, V_0 = 0.481 × E µmol min^{-1}, described the data with R^2 = 0.972, and a free intercept was indistinguishable from zero (p = 0.98; Figure 2B). The trend is approximately but not exactly proportional: the rate per unit enzyme ranged from 0.71 (tube D) to 1.21 (tube C), and tubes D and E (0.25 and 0.15) gave nearly equal rates.

β-Galactosidase approaches saturation with ONPG

Without inhibitor, only the 2.20 mM tube remained linear to 10 min (V_0 = 0.132 ΔA_{415} min^{-1}); in tubes B–F the later increments fell below the first, so no linear window could be fitted and V_0 was taken as the origin-assumed 0–2 min average rate, A_{415}(2 min)/2 min with A_{415} = 0 assumed at t = 0 (†; Figure 3A; Tables S2 and S3). V_0 rose 5.6-fold, from 0.081 to 0.455 µmol min^{-1}, over the 32-fold substrate range and flattened above 1 mM (Figure 4A). The Lineweaver–Burk regression, $1/V_0$ = 0.658(1/[S]) + 2.097 (R^2 = 0.928; Figure 4B), gave V_{max} = 0.477 µmol min^{-1} and K_m = 0.31 mM; direct nonlinear least squares on the velocity scale gave K_m = 0.36 ± 0.13 mM and V_{max} = 0.494 ± 0.059 µmol min^{-1} (Table 1). As a sensitivity scenario, setting the t = 0 absorbance of those tubes to the median intercept of the curves that were linear (0.013 A; range -0.020 to 0.059 A) lowers each 0–2 min estimate by about 0.006 ΔA_{415} min^{-1} and moves K_m to 0.55 mM (Lineweaver–Burk) or 0.52 mM (nonlinear) (Table S5); intercepts of other reactions do not bound the baseline of these tubes, so the scenario is a what-if calculation, not an uncertainty interval.

IPTG raises the apparent K_m and inhibits most at low substrate

With 0.458 mM IPTG, every tube was slower (Figure 3B): by 25% at 2.20 mM ONPG, 68% at 1.10 mM, and 87% at 0.069 mM, so inhibition grew as substrate fell (Figure 4A; Table S3). The Lineweaver–Burk line steepened 9.5-fold, $1/V_0$ = 6.24(1/[S]) + 3.55 (R^2 = 0.988), giving $K_{m,app}$ = 1.76 mM (5.6-fold increase) and $V_{max,app}$ = 0.282 µmol min^{-1} (Figure 4B; Table 1). The inhibited velocities halved with each halving of substrate from tube B downward and no plateau was reached within the tested range, so $V_{max,app}$ rests on an extrapolated intercept (3.55 ± 2.33) and a nonlinear fit of the inhibited series alone ran to the K_m bound. The independently fitted $V_{max,app}$ is 0.59 of the uninhibited V_{max}, but the data cannot establish whether V_{max} changed. The competitive relation gives K_i = [I]/($K_{m,app}/K_m$ − 1) = 0.458/(1.76/0.31 − 1) = 0.100 mM.

To compare mechanisms on equal terms, three models were fitted to both series at once (Table 1; Table S6). A competitive model (shared V_{max}, K_m raised by 1 + [I]/K_i) gave V_{max} = 0.53 ± 0.07 µmol min^{-1}, K_m = 0.44 ± 0.17 mM, and K_i = 0.107 ± 0.042 mM (Figure 4A, solid curves). A pure noncompetitive model with the same number of parameters (shared K_m, V_{max} reduced by 1 + [I]/K_i) left a residual sum of squares 1.9 times larger (ΔAIC = 7.8), and a mixed model reduced to the competitive one (its uncompetitive constant ran to the bound of 500 mM). The equal V_{max} of the competitive and mixed fits is an assumption of those models rather than an observation; what the data show is that the competitive model describes the twelve points better than the noncompetitive alternative.

Our own measurements reproduce the pattern

The Results above use the reference dataset. Our experimental dataset, recorded on the same instrument and analyzed with the identical rule and units, gave the same picture (Figures S1 and S4; Table S7). The standard tube agreed with the reference series within 5% (enzyme dilution) and 15% (substrate series); nonlinear fits gave K_m = 0.14 mM and V_{max} = 0.41 µmol min^{-1} without inhibitor, and IPTG raised K_m 8.8-fold while V_{max} was unchanged (ratio 1.01), with K_i = 0.059 mM. Two tubes deviated. The inhibited tube nominally containing 1.10 mM ONPG ran 1.57-fold faster than tube A at half the substrate, which no kinetic mechanism permits at fixed enzyme, and 3.7-fold faster than the reference tube B; its profile matches an uninhibited high-substrate tube, consistent with IPTG omitted or ONPG or enzyme over-pipetted (hypotheses, not findings). Its values are retained as recorded and the tube is excluded from fits; including it gives $K_{m,app}$ = 38 mM. The inhibited 0.1375 mM tube was discarded before measurement.

The conclusions that survive alternative estimators, and those that do not

Because the choice of initial-rate window drives K_m, every parameter was recomputed under five estimators (Figure S3; Table S5): all five points of every curve, the initial-window regression for every curve, the origin-assumed 0–2 min average rate for every curve, the rule under the baseline-offset scenario, and the rule as used. Regression alone gives a negative Lineweaver–Burk intercept for the uninhibited series (non-physical), and all five points give K_m = 3.4 mM, because both average over the falling phase of the curved low-substrate tubes; the 0–2 min average rate applied to every curve collapses $K_{m,app}$ to 0.62 mM and raises K_i to 0.50 mM, because for the inhibited tubes, which were linear over 2–10 min, it replaces a five-point regression with a single interval; the offset scenario moves K_i to 0.22 mM. Saturation with substrate and inhibition that grows as substrate falls survive every estimator; the numerical K_m (0.31–0.55 mM among the rule-based estimators and the global fit) and K_i (0.10–0.22 mM among those; 0.50 mM under the uniform 0–2 min estimate) do not; these are alternative method-dependent estimates, not confidence intervals.

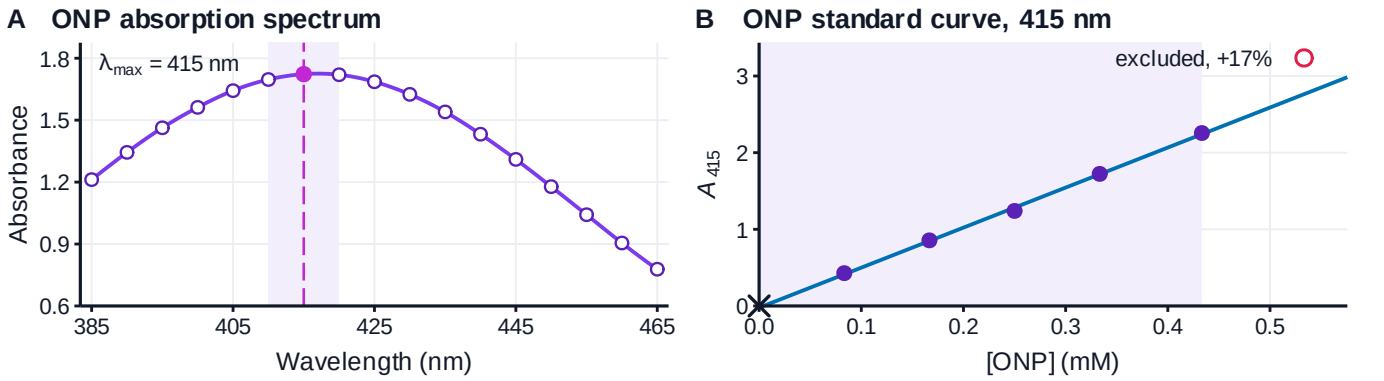


Figure 1. *o*-nitrophenol absorbs maximally at 415 nm and obeys the Beer–Lambert law to 0.43 mM
(A) Absorbance spectrum of 0.333 mM ONP (1.0 mL of 1 mM ONP + 1.0 mL Z buffer + 1.0 mL 1 M Na_2CO_3) read from 465 to 385 nm in 5 nm steps against the Z buffer/Na_2CO_3 blank. Points are readings, the line a spline, the dashed line the maximum at 415 nm (labeled); the shaded band (410–420 nm) lies within 2% of the maximum.
(B) Standard curve at 415 nm for 1 mM ONP diluted to 0.083–0.533 mM in the same 3.0 mL matrix. Filled points and the line are the least-squares calibration over the linear region (shaded): A_{415} = 5.222[ONP] − 0.022, R^2 = 0.9988, n = 5, ε = 5.22 ± 0.10 mM^{-1} cm^{-1}. ×, zero-concentration point (not fitted). The 0.533 mM standard (open, labeled) lay 17% above the line and was excluded. Reference dataset, Part I; see STAR Methods.

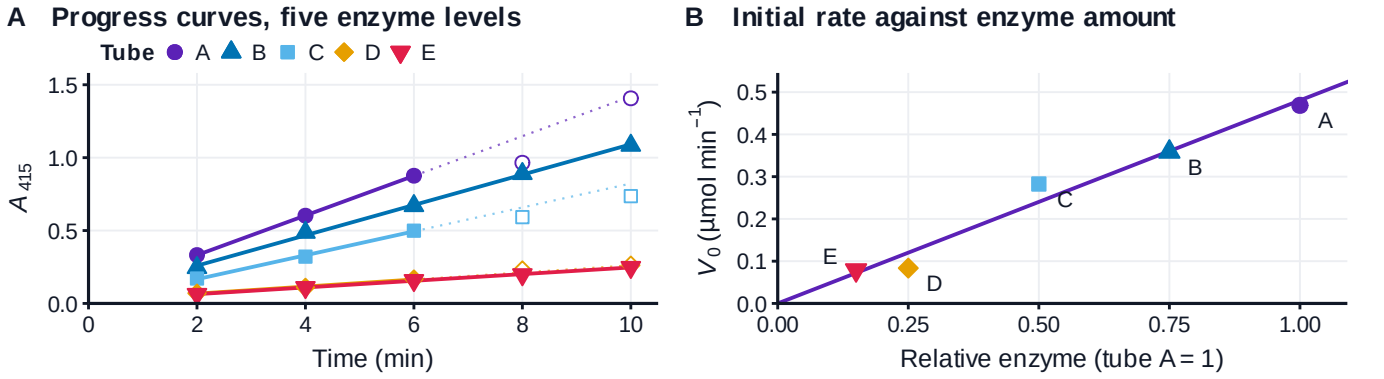


Figure 2. Initial velocity rises approximately in proportion to the amount of β-galactosidase
(A) Progress curves (A_{415} of 2.0 mL aliquots stopped with 1.0 mL 1 M Na_2CO_3) for 2.20 mM ONPG hydrolyzed by 1.00, 0.75, 0.50, 0.25, and 0.15 relative amounts of the standardized enzyme (tubes A–E). Lines are least-squares fits to the initial linear window chosen by the operational rule (STAR Methods); filled points were used, open points (8 min of tubes A and C, below the initial trend) were excluded, dotted lines extrapolate; V_0 values are given in the text.
(B) V_0 in µmol ONP min^{-1} per 12.0 mL assay (× 3.447 from ΔA_{415} min^{-1}) against relative enzyme; letters identify the tubes. Line, proportional model through the origin, V_0 = 0.481 × E (R^2 = 0.972); a free-intercept fit (intercept -0.0006 µmol min^{-1}, p = 0.98) coincides with it and is not drawn. n = 1 curve per tube; reference dataset, Part II.

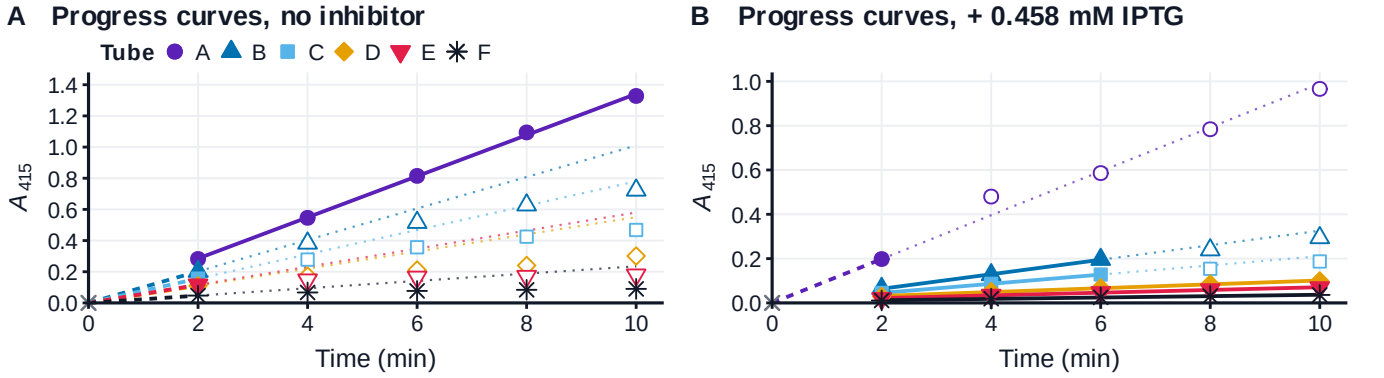


Figure 3. Product formation slows as substrate falls and when IPTG is present
(A) Progress curves for 0.2 mL enzyme acting on 2.20, 1.10, 0.55, 0.275, 0.1375, and 0.069 mM ONPG in 12.0 mL Z buffer at room temperature (tubes A–F, no inhibitor). Only tube A stayed linear to 10 min (regression over the filled points); in tubes B–F the later increments fell below the first, so V_0 is the origin-assumed 0–2 min average rate (dashed segment from ×, where A_{415} = 0 is assumed at t = 0 because no reading exists), an average over the first interval rather than a measured instantaneous rate; later points (open) were not used; dotted lines extrapolate the fits.
(B) The same six tubes with 0.458 mM IPTG (1.0 mL of 5.5 mM IPTG per 12.0 mL), on a smaller absorbance scale. Tube A is origin-assumed (its 4–10 min increments fell below the first); tubes B–F were fitted by regression over the filled points. V_0 values are listed in Table S3. n = 1 curve per tube; reference dataset, Part III.

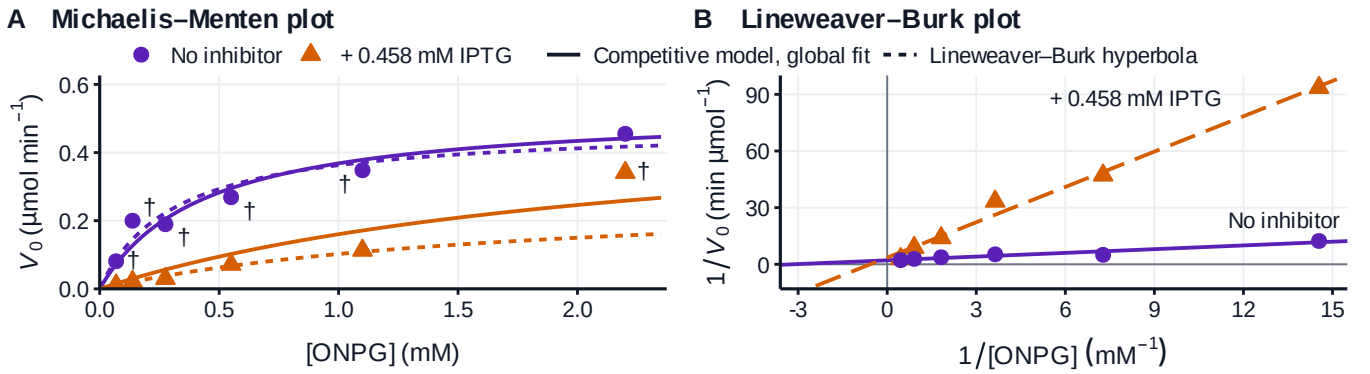


Figure 4. β-Galactosidase approaches saturation with ONPG, and IPTG raises the apparent K_m
(A) V_0 against [ONPG] without inhibitor (circles) and with 0.458 mM IPTG (triangles), from the progress curves in Figure 3 (tubes A–F). Solid curves, a single competitive-inhibition model fitted to both series at once; dashed curves, hyperbolae implied by the two Lineweaver–Burk lines in (B) (shared V_{max} = 0.53 ± 0.07 µmol min^{-1}, K_m = 0.44 ± 0.17 mM, K_i = 0.107 ± 0.042 mM; the shared V_{max} is a model assumption). † marks origin-assumed 0–2 min estimates.
(B) Double-reciprocal plot of the same velocities. No inhibitor: $1/V_0 = 0.66(1/[S]) + 2.10$, R^2 = 0.928, giving V_{max} = 0.477 µmol min^{-1} and K_m = 0.31 mM. + IPTG: $1/V_0 = 6.24(1/[S]) + 3.55$, R^2 = 0.988, giving $V_{max,app}$ = 0.282 µmol min^{-1} and $K_{m,app}$ = 1.76 mM. x-intercepts: $-1/K_m$ = -3.19 and $-1/K_{m,app}$ = -0.57 mM^{-1}. n = 1 curve per point; reference dataset, Part III.

DISCUSSION

What the reference measurements demonstrate directly is that the velocity of β-galactosidase rose with the amount of enzyme and saturated with ONPG, and that IPTG inhibited the enzyme most where substrate was lowest, raising the apparent K_m 5.6-fold. What rests on a model is the mechanism: a competitive model described both series better than a pure noncompetitive one with the same number of parameters, and letting IPTG also bind the enzyme–substrate complex gained nothing, so the behaviour is consistent with competitive inhibition. It is not a proof. The inhibited series never approached saturation, so its independently fitted $V_{max,app}$ (0.59 of the uninhibited value) is an extrapolation that cannot show whether V_{max} changed, and the competitive model shares V_{max} by construction, so its fit cannot show that V_{max} was unchanged. K_i is 0.10 mM from the Lineweaver–Burk pair, 0.11 mM from the global fit, 0.22 mM under the baseline-offset scenario and 0.50 mM if every curve is reduced to its 0–2 min average rate: alternative method-dependent estimates rather than a confidence interval, which place K_i between 0.1 and 0.5 mM under the tested conditions; the 0.08 mM measured for the purified enzyme with ONPG in TES buffer with 145 mM NaCl and 0.01 mM Mg^{2+} at pH 7.0 and 25 °C[10] is of the same order, under conditions (10 mM K^+, 1 mM Mg^{2+} in Z buffer here) that were not expected to give identical values. The structural picture predicts this behaviour: IPTG occupies the galactose subsite as a dead-end thiogalactoside and is displaced by substrate.[6]

K_m (0.31–0.55 mM) is two to five times the 0.12 mM reported for the purified enzyme,[10] and several factors may contribute: five of the six uninhibited V_0 values are origin-assumed 0–2 min average rates, which understate the initial rate if it was already falling within that interval and so inflate $1/V_0$ at low [S] and K_m; the assay conditions (Z buffer, ambient temperature) differ from the literature assay; 2-min manual sampling limits the resolution of fast early phases; and the effective ONPG concentrations in the most dilute tubes carry pipetting and stock uncertainty. With 2-min sampling, the kinetic constants of a fast enzyme are only as good as the rule used to define the initial rate, which is why we state that rule, examine its origin assumption and report every alternative. The deceleration itself is not explained. Those curves flattened at about 38% of the nominal product yield in tubes E and F (Table S4), which substrate depletion alone does not explain; the timing of the first aliquots, reaction preparation, a lower-than-nominal effective ONPG concentration, stopping and read-out effects could each contribute, and the data do not single one out. The ONPG hypothesis is testable by measuring the stock's own absorbance or by complete hydrolysis of an aliquot. The calibration was not the limiting step: it was linear over five standards (R^2 = 0.9988); the one excluded standard read 0.06% transmittance near the instrument's linear limit and deviated upward, so its cause is undetermined, and its inclusion would have changed V_{max} by 13% at most and K_m and K_i not at all. The standards and the stopped samples share the alkaline matrix, but the kinetic blank was Z buffer alone; the small positive intercepts of the linear curves measure any residual mismatch.

Our experimental dataset reproduced the kinetic pattern with

Table 1. Kinetic parameters of β-galactosidase for ONPG without and with 0.458 mM IPTG
Reference dataset, tubes A–F in every fit; ± values are standard errors of the fit (single measurements, no replicates). Lineweaver–Burk is the course method, the nonlinear fits a check; app., apparent value with IPTG; n.d., not determinable (no plateau within 0.069–2.2 mM). V_{max} in µmol ONP min^{-1} per 12.0 mL assay. K_i from $K_{m,app} = K_m(1 + [I]/K_i)$ for the Lineweaver–Burk pair and from models fitted to both series with a shared V_{max} (competitive) or a shared K_m (noncompetitive); RSS, residual sum of squares. –, not applicable.

Condition	Method	K_m (mM)	V_{max} (µmol min^{-1})	K_i (mM)	Fit
No inhibitor	Lineweaver-Burk	0.314	0.477	–	R2 0.928
No inhibitor	Nonlinear MM	0.363 ± 0.128	0.494 ± 0.059	–	R2 0.919
+ 0.458 mM IPTG	Lineweaver-Burk	1.757 (app.)	0.282 (app.)	0.100	R2 0.988
+ 0.458 mM IPTG	Nonlinear MM	n.d.	n.d.	–	R2 0.962
Both series	Global, competitive	0.435 ± 0.166	0.529 ± 0.072	0.107 ± 0.042	RSS 0.0194
Both series	Global, noncompetitive	0.744 ± 0.382	0.626 ± 0.139	0.433 ± 0.181	RSS 0.0371

a K_m (0.14 mM) closer to the literature value, because our low-substrate tubes stayed linear longer than the reference tubes. Its two deviating tubes illustrate the value of recording everything: the anomalous inhibited tube B is reported with its raw values and excluded only from the fits, with the consequence of including it quantified, and the discarded tube E is left missing rather than estimated. Improvements would be 30-s sampling or continuous monitoring during the first 2 min, substrate concentrations from 0.03 mM to well above 2 mM so that both series approach saturation, duplicate tubes, and spectrophotometric verification of the ONPG stock.

Biologically, the inducer of the *lac* operon also competes with substrates at the active site of the enzyme it induces, and the same principle, a stable substrate analog that raises the substrate requirement without lowering catalytic capacity, defines competitive drugs from statins to protease inhibitors.

Limitations of the study

Each condition was measured once, so no replicate variance is available and the standard errors quoted come from single regressions. Manual 2-min sampling cannot resolve the initial rate when it falls within the first interval, which affected five of six uninhibited reference tubes; the 0–2 min estimates for those tubes rest on an assumed zero absorbance at t = 0 that was examined in a sensitivity scenario but cannot be verified from these data. The substrate range reached only 2.2 mM, below the apparent K_m with IPTG, so the inhibited plateau was never observed and $V_{max,app}$ is extrapolated; the mechanism is inferred from model comparison on twelve single measurements. Read-out resolution (0.001 *A*) is coarse relative to the 0.005 *A* increments of the most dilute inhibited tubes. The plateaus of the low-substrate reference curves suggest that nominal ONPG concentrations may have been lower than stated. Temperature was ambient and not controlled. Literature values were measured in a different buffer by a continuous assay, so exact agreement is not expected.

STAR★METHODS

Key resources table

See Table 2.

Resource availability

Lead contact Further information and requests should be directed to the lead contact, Almokhtar Aljarodi (almokhtaraljarodi@gmail.com).

Materials availability This study did not generate new unique reagents.

Data and code availability All raw absorbance values (the reference dataset as transcribed from the course's files, and our 14 printouts transcribed and verified value by value), the processed data, every table and figure, and the R scripts that regenerate them are in the public repository `https://github.com/almokhtar8-stack/exp2`. `Rscript R/run_all.R` reproduces every number in this manuscript; the text itself is filled from the processed outputs so that prose and data cannot disagree. Any additional information is available from the lead contact upon request.

Experimental model and study participant details

No organisms, cell lines, or human participants were used. The enzyme was *Escherichia coli* β-galactosidase (EC 3.2.1.23) supplied by the course as a solution standardized to give A_{415} = 1.20–1.30 after 10 min in the standard reaction (0.2 mL enzyme, 2.20 mM ONPG, 12.0 mL).

Method details

Datasets Two datasets were analyzed: a course-provided reference dataset supplied by the laboratory instructor (Professor Samah Gadhoum; Parts I/II on 10 September 2026 and Part III on 13 September 2026, with the instruction to use them for the Results section) and an experimental dataset collected by our laboratory group on 7 September 2026 (Part II, tubes A–E) and 11 September 2026 (Part III, tubes A–E without IPTG, A–D with IPTG; tube F not performed, inhibited tube E discarded before measurement). The reference measurements were used for the primary kinetic analysis; our experimental measurements were analyzed identically and examined separately for technical interpretation and troubleshooting. The labels describe origin and analytical role, not accuracy or publication status, and the two datasets are not replicates of each other. Reference values were transcribed from the course's .docx and .pptx files and cross-checked value by value against the course's CSV files; the enzyme-dilution table absent from the CSV was transcribed from the .docx. No missing value was estimated.

Reagents Z buffer contained 0.06 M Na_2HPO_4, 0.04 M NaH_2PO_4, 0.01 M KCl, 0.001 M $MgSO_4$, and 0.05 M β-mercaptoethanol, pH 7.0. Stocks were ONPG at 1.0 mg mL^{-1} (3.3 mM), IPTG at 1.31 mg mL^{-1} (5.5 mM), ONP at 1 mM, and Na_2CO_3 at 1 M. All procedures followed the Experiment 2 laboratory manual[11] except as noted.

Absorption spectrum and standard curve The spectrum of 1.0 mL of 1 mM ONP in 1.0 mL Z buffer and 1.0 mL 1 M Na_2CO_3 (0.333 mM ONP) was read from 465 to 385 nm in 5 nm steps against the blank (2.0 mL Z buffer + 1.0 mL Na_2CO_3). Standards contained 0.25, 0.50, 0.75, 1.0, 1.3, and 1.6 mL of 1 mM ONP made up to 2.0 mL with Z buffer plus 1.0 mL 1 M Na_2CO_3 ([ONP] = 1 mM × *V*/3.0 mL) and were read at 415 nm; the manual's 2.0 mL standard has no reported reading (Table S1).

Enzyme dilution Five 12.0 mL reactions contained 8.0 mL of 3.3 mM ONPG (2.20 mM), Z buffer, and 0.20, 0.15, 0.10, 0.05, or 0.03 mL of enzyme (relative enzyme 1.00, 0.75, 0.50, 0.25, 0.15), enzyme added last to start the timer.

Substrate series and inhibition Twelve 12.0 mL reactions contained 8.0, 4.0, 2.0, 1.0, 0.5, or 0.25 mL of 3.3 mM ONPG ([ONPG] = 2.20, 1.10, 0.55, 0.275, 0.1375, or 0.06875 mM; tubes A–F), Z buffer to volume, 0.2 mL enzyme, and either 1.0 mL Z buffer or 1.0 mL of 5.5 mM IPTG ([IPTG] = 0.458 mM).

Stopped-reaction sampling At 2, 4, 6, 8, and 10 min after enzyme addition, 2.0 mL aliquots were transferred to tubes containing 1.0 mL 1 M Na_2CO_3, which stops the reaction and develops the *o*-nitrophenolate color, and A_{415} was read against a Z-buffer blank in a GENESYS 10S UV-Vis spectrophotometer (1 cm path) at room temperature.

Quantification and statistical analysis

Concentrations Reaction concentrations were calculated from stock concentrations and volumes: [ONPG] = 3.3 mM × *V*/12.0 mL; [IPTG] = 5.5 mM × 1.0/12.0 mL; ONP standards 1 mM × *V*/3.0 mL. Every reaction sums to 12.0 mL and every standard to 3.0 mL (checked programmatically).

Table 2. Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
β-Galactosidase (*E. coli*), standardized solution	B241 teaching laboratory	N/A
o-Nitrophenyl-β-D-galactopyranoside (ONPG), 3.3 mM	B241 teaching laboratory	N/A
Isopropyl-β-D-thiogalactopyranoside (IPTG), 5.5 mM	B241 teaching laboratory	N/A
o-Nitrophenol (ONP), 1 mM	B241 teaching laboratory	N/A
Sodium carbonate, 1 M	B241 teaching laboratory	N/A
Z buffer (phosphate, KCl, $MgSO_4$, β-mercaptoethanol; pH 7.0)	B241 teaching laboratory	N/A
Deposited data		
Reference dataset (course-provided Part I and Part III measurements)	B241 teaching staff (e-mail of 13 Sep 2026)	`data/raw/reference/`
Experimental dataset (our 14 instrument printouts, transcribed)	This report	`data/raw/experimental/`; `report/raw_data_printouts/`
Processed data, tables, figures	This report	`data/processed/`, `tables/`, `figures/`
Software and algorithms		
R 4.3.3	R Core Team	https://www.r-project.org
tidyverse	Wickham et al.[12]	https://www.tidyverse.org
broom, patchwork, scales, here	CRAN	https://cran.r-project.org
Analysis scripts (`R/00_setup.R` to `R/09_graphical_abstract.R`, `R/run_all.R`)	This report	Repository, `R/`
Other		
GENESYS 10S UV-Vis spectrophotometer	Thermo Fisher Scientific[13]	N/A

Initial velocities The manual asks for V_0 from the initial linear portion of each progress curve, with the points used identified. One rule, depending only on the shape of each curve, was applied to all 31 progress curves. Tier 1: the least-squares line $A_{415} = a + bt$ was fitted to the 2, 4, and 6 min points and extended one point at a time while the next increment lay within 20% (or 0.010 *A*, whichever is larger) of the increment predicted by the current line; $V_0 = b$. Tier 2: when the second and third increments both fell below the first by more than this tolerance, no linear window could be fitted within the sampled interval, and V_0 was taken as the average rate over the first interval, A_{415}(2 min)/2 min, which assumes $A_{415} = 0$ at t = 0 because no reading exists there; this origin-assumed estimate is an average over 0–2 min rather than a measured instantaneous initial velocity, may lie below or above the true V_0 depending on the unmeasured early course, and is flagged (†) wherever it appears. Tier 2 applied to 6 of the 17 reference curves (uninhibited B–F, inhibited A) and to one of our own curves. As a sensitivity scenario, the intercepts of the 24 curves that were linear (median 0.013 A, range -0.020 to 0.059 A) were used as an assumed t = 0 absorbance for the origin-assumed curves, and this scenario estimator was carried through every analysis; because those intercepts belong to other reactions, the scenario is an assumption transferred between curves, not a measured baseline or a bound. The extinction coefficient was the slope of A_{415} against [ONP] over the linear region chosen by the same increment rule.

Units V_0 in ΔA_{415} min^{-1} was converted to µmol ONP min^{-1} per assay as $V_0/\varepsilon \times 1.5$ (2 mL reaction + 1 mL stop reagent) × 12.0 mL (1 ΔA_{415} min^{-1} = 3.447 µmol min^{-1} per assay); the manual's instruction to divide the slope by ε gives the cuvette rate in mM min^{-1}, and the two volume factors convert it to an amount per minute. K_m and K_i do not depend on this factor; V_{max} scales with it.

Kinetic models K_m and V_{max} were obtained from the Lineweaver–Burk regression (V_{max} = 1/intercept, K_m = slope × V_{max}) and by unweighted nonlinear least squares of $V_0 = V_{max}[S]/(K_m + [S])$ on the velocity scale (`nls`, bounded "port" algorithm from a grid of starting values; a fit that runs to the K_m bound of 100 mM is reported as "no plateau within the tested range"). K_i was calculated from $K_{m,app} = K_m(1 + [I]/K_i)$. Three global models were fitted to both series simultaneously: competitive, $V = V_{max}[S]/(K_m(1 + [I]/K_i) + [S])$; pure noncompetitive, $V = V_{max}[S]/((K_m + [S])(1 + [I]/K_i))$; and mixed, $V = V_{max}[S]/(K_m(1 + [I]/K_i) + [S](1 + [I]/K_i'))$; the first two have the same number of parameters and were compared by residual sum of squares and AIC. Reported uncertainties are regression standard errors from single measurements; no replicate statistics are reported.

Software All calculations were performed in R 4.3.3 with tidyverse,[12] broom, patchwork, scales, and here; figures were drawn with ggplot2 at 600 dpi. The scripts, raw data, and outputs are in the repository listed under data and code availability.

SUPPLEMENTAL INFORMATION

Supplemental information includes Figures S1–S4 and Tables S1–S7 (following the references), the reference diagram for the course's Question 1, and the 14 instrument printouts (repository, `report/raw_data_printouts/`).

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AUTHOR CONTRIBUTIONS

Investigation, A.A. and A.K.; formal analysis, software, visualization, and writing, A.A.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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SUPPLEMENTAL INFORMATION

Figures S1–S4 and Tables S1–S7 supporting the main text. All items are regenerated from the raw data by `Rscript R/run_all.R`.

Table S1. Mapping of the reported standard-curve readings to the manual's preparations, related to Figure 1
The course table lists seven readings (tubes 1–7); the manual describes a blank and seven standards. The first reading is the zero-concentration point (confirmed by the instructor); the remaining six follow the manual's standards 1–6, anchored by the spectrum tube. The manual's standard 7 has no reported reading; whether it was unprepared or unreported is unknown.

Reported tube	Manual preparation	ONP 1 mM (mL)	Z buffer (mL)	Na_2CO_3 (mL)	[ONP] (mM)	A_{415}	Basis
1	Blank (0 mL ONP)	0	2	1	0	0.000	Reported as tube 1 with OD 0; the instructor confirmed it is the zero-concentration point (blank-like preparation, read against itself)
2	Standard 1	0.25	1.75	1	0.0833	0.428	Next reported reading; mapping consistent with Beer-Lambert ratio A/[ONP] = 5.1 per mM
3	Standard 2	0.5	1.5	1	0.1667	0.856	A/[ONP] = 5.1 per mM
4	Standard 3	0.75	1.25	1	0.25	1.240	A/[ONP] = 5.0 per mM
5	Standard 4	1	1	1	0.3333	1.723	Identical to the spectrum reading of manual tube 4 at 415 nm (1.723), which anchors the mapping
6	Standard 5	1.3	0.7	1	0.4333	2.256	A/[ONP] = 5.2 per mM
7	Standard 6	1.6	0.4	1	0.5333	3.236	A/[ONP] = 6.1 per mM; highest reported reading
not reported	Standard 7	2	0	1	0.6667	none	Not in the reported table (0.667 mM); unknown whether this standard was not prepared or not reported

Figure S1 Progress curves of the two datasets on common axes

Solid segments: regression windows used for V_0; dashed: origin-assumed 0–2 min average rate (no reading at t = 0); open: not used for V_0

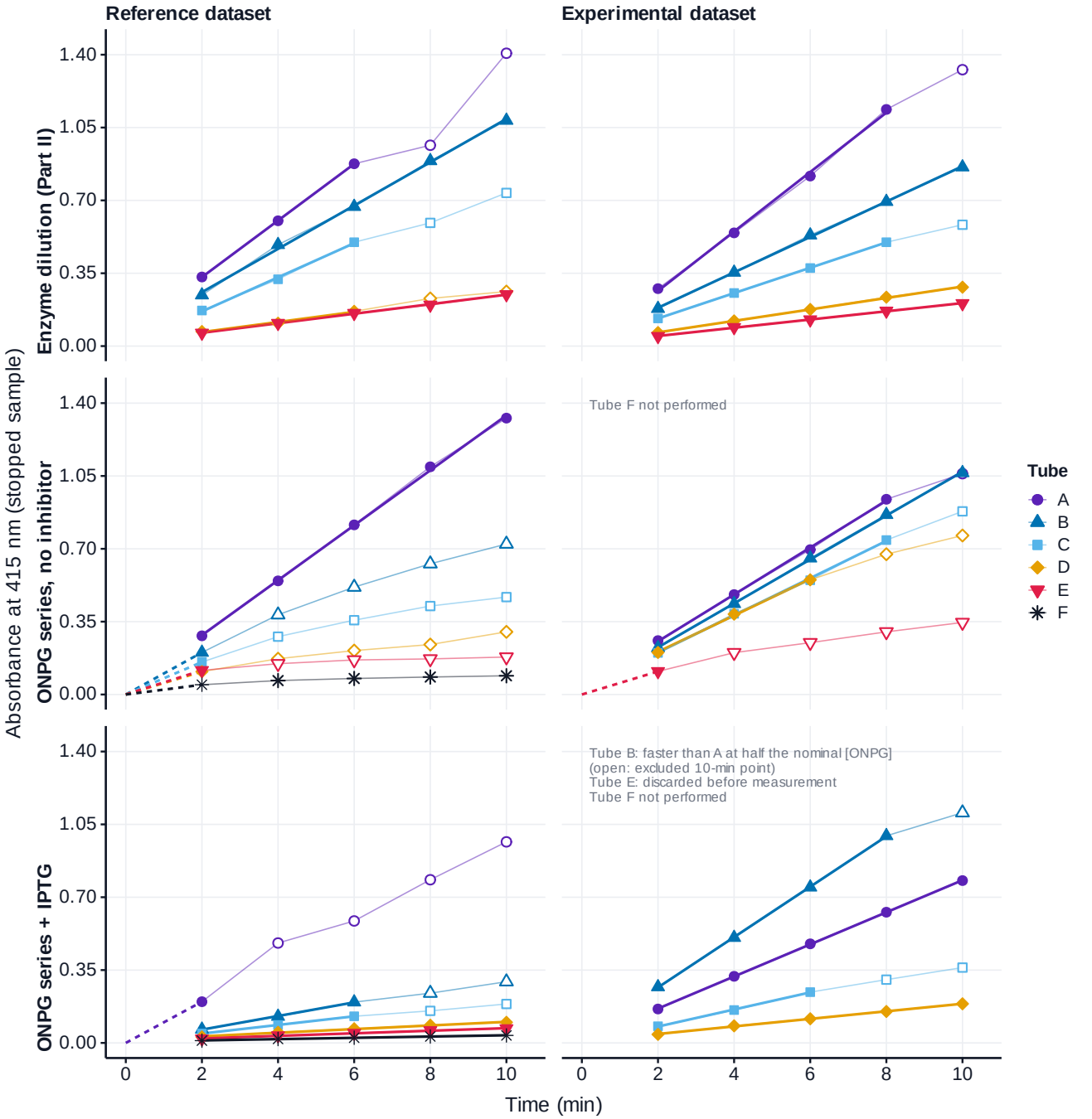


Figure S1. Progress curves of the two datasets on common axes, related to Figures 2–4

Reference dataset (left column) and experimental dataset (right column; our 14 GENESYS 10S printouts, 7 and 11 September 2026) for the three experiments. Solid segments are the regression windows and dashed segments the origin-assumed 0–2 min average rates used for V_0 under the identical rule. The standard tube (A: 0.2 mL enzyme, 2.20 mM ONPG) agreed between the two datasets within 5% (enzyme dilution) and 15% (substrate series). Our low-substrate tubes without inhibitor stayed linear longer than the reference tubes (only tube E was non-linear from the outset), and our inhibited tubes C and D ran about twice as fast as the reference tubes. Our inhibited tube B (open) ran faster than tube A at half the nominal substrate and is excluded from kinetic fits; our inhibited tube E has no data (sample discarded); tube F was not performed on either day.

Figure S2 Successive 2-min increments relative to the fitted initial velocity

Increment / (V_0 × 2 min); shaded: ±20% tolerance of the window rule (or ±0.010 A); open: intervals not used for V_0
For origin-assumed curves V_0 is the 0–2 min average rate, so every later interval falls below 1

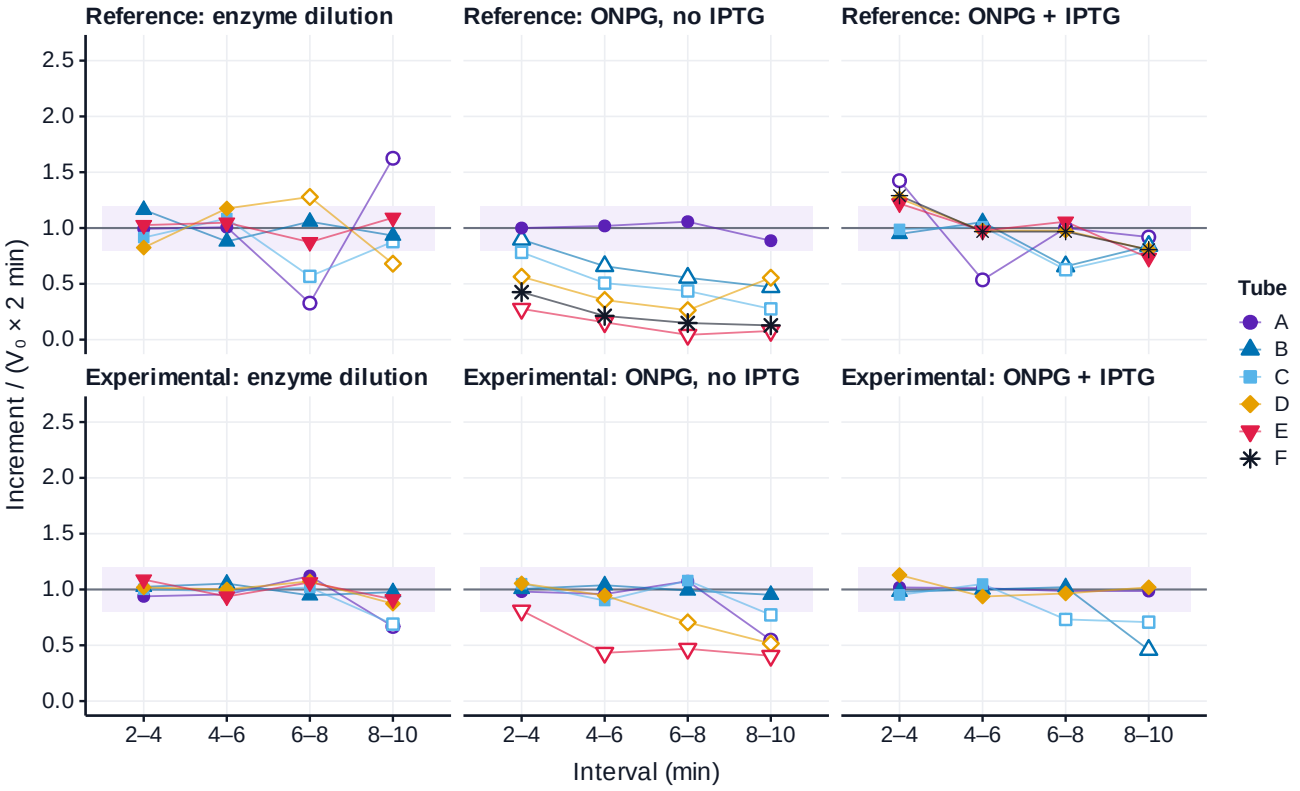


Figure S2. Increment diagnostics behind every V_0 decision, related to STAR Methods
For each of the 31 measured curves, every successive 2-min increment divided by the increment predicted by the fitted V_0 (V_0 × 2 min). A value of 1 means the interval ran at the fitted initial rate; the shaded band is the acceptance window of the rule (±20% or ±0.010 *A*, whichever is larger). Open symbols are intervals not used for V_0. For curves with no initial linear portion (reference no-IPTG B–F, reference +IPTG A, experimental no-IPTG E) the reference V_0 is the origin-assumed 0–2 min average rate, so all later intervals fall below 1, showing how the measured increments declined: to 35–60% of the first-interval rate by the second interval in the reference uninhibited tubes B–F.

Figure S3 Kinetic parameters of the reference dataset under five V_0 estimators

Log scale. Nonlinear fits of the inhibited series ran to the K_m bound (no plateau within 0.069–2.2 mM): no $K_{m,app}$ or K_i for that method
Cyan: values for purified E. coli β-galactosidase (TES buffer, pH 7.0, 25 °C)

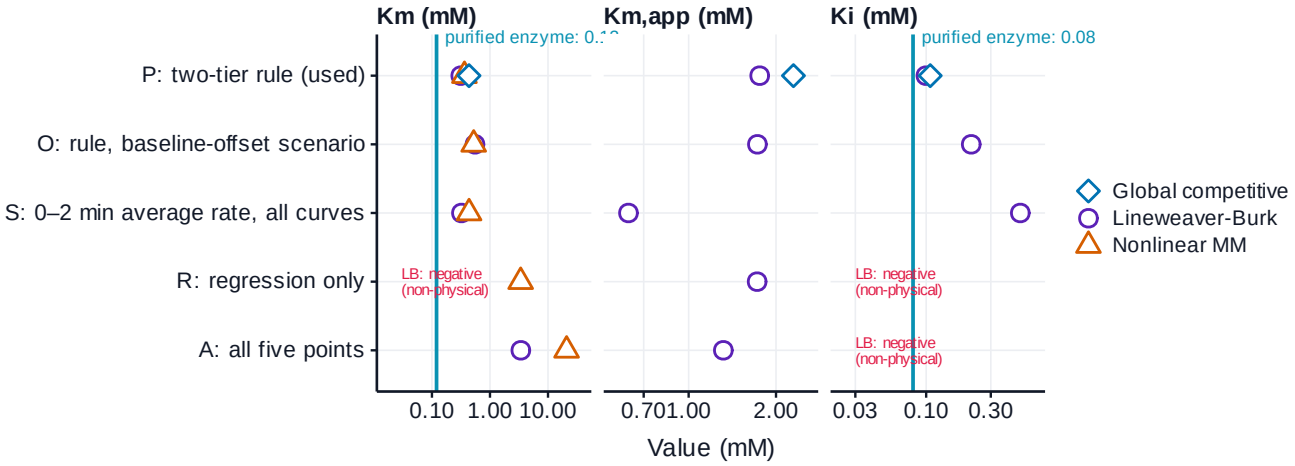


Figure S3. Sensitivity of the kinetic parameters to the initial-velocity estimator, related to Figures 3 and 4

K_m (no inhibitor), $K_{m,app}$ (+IPTG), and K_i from the reference dataset under five ways of estimating V_0: A, all five points of every curve; R, the initial-window regression for every curve; S, the origin-assumed 0–2 min average rate for every curve; O, the two-tier rule under the baseline-offset scenario (t = 0 absorbance set to the median intercept of the linear curves, 0.013 A, an assumption transferred between reactions); P, the two-tier rule used in the Results. Circles, Lineweaver–Burk; triangles, nonlinear Michaelis–Menten; diamond, global competitive fit. With regression-only V_0 the uninhibited Lineweaver–Burk line has a negative intercept (non-physical K_m and V_{max}, noted in red), and with all five points K_m exceeds 3 mM, because both estimators average over the falling phase of the curved low-substrate tubes. The nonlinear +IPTG fits found no plateau within the tested range under any estimator. Cyan lines, K_m = 0.12 mM and K_i = 0.08 mM reported for purified *E. coli* β-galactosidase with ONPG (TES buffer, pH 7.0, 25 °C).[10]

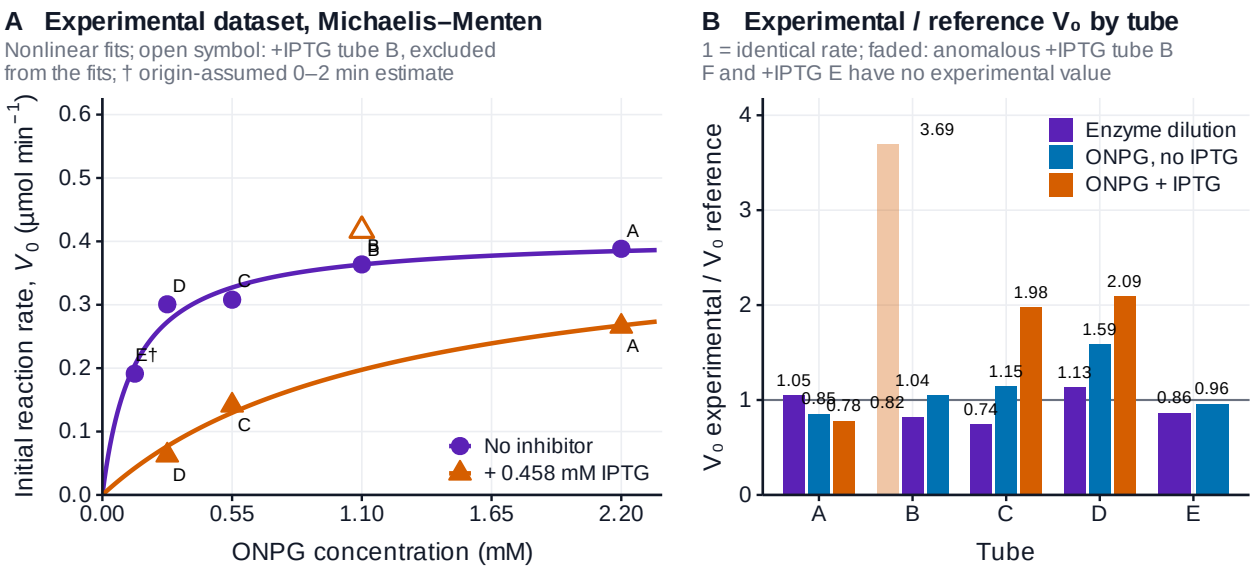


Figure S4. Kinetic analysis of the experimental dataset, related to Figure 4
(A) Michaelis–Menten representation of our Part III data with the same V_0 rule and units as the Results, with the anomalous inhibited tube B (open) excluded and the inhibited tube E absent. Nonlinear fits: K_m = 0.14 mM, V_{max} = 0.41 µmol min^{-1} without inhibitor; $K_{m,app}$ = 1.21 mM, $V_{max,app}$ = 0.41 with IPTG (ratio 1.01); K_i = 0.059 mM.
(B) Ratio of our V_0 to the reference V_0 for every tube measured in both datasets. The standard tube agreed within 5–15%; our uninhibited tube D and inhibited tubes C and D ran 1.6- to 2.1-fold faster; the anomalous inhibited tube B (faded) ran 3.7-fold faster than the reference tube B.

Table S2. Primary absorbance data (A_{415}) for the substrate series without and with IPTG, related to Figures 3 and 4
Reference dataset, Part III. Reaction concentrations from the manual's volumes ([ONPG] = 3.3 mM × V/12.0 mL; [IPTG] = 5.5 mM × 1.0/12.0 mL). Absorbance of 2.0 mL aliquots stopped with 1.0 mL 1 M Na_2CO_3, read at 415 nm.

Condition	Tube	ONPG (mL)	[ONPG] (mM)	[IPTG] (mM)	2 min	4 min	6 min	8 min	10 min
ONPG series, no inhibitor	A	8	2.2	0	0.282	0.546	0.815	1.094	1.328
ONPG series, no inhibitor	B	4	1.1	0	0.202	0.383	0.516	0.628	0.723
ONPG series, no inhibitor	C	2	0.55	0	0.156	0.278	0.357	0.425	0.468
ONPG series, no inhibitor	D	1	0.275	0	0.11	0.172	0.211	0.24	0.301
ONPG series, no inhibitor	E	0.5	0.1375	0	0.116	0.148	0.166	0.171	0.18
ONPG series, no inhibitor	F	0.25	0.06875	0	0.047	0.067	0.077	0.084	0.09
ONPG series + IPTG	A	8	2.2	0.458	0.198	0.48	0.586	0.784	0.966
ONPG series + IPTG	B	4	1.1	0.458	0.065	0.127	0.196	0.239	0.294
ONPG series + IPTG	C	2	0.55	0.458	0.045	0.086	0.128	0.154	0.187
ONPG series + IPTG	D	1	0.275	0.458	0.029	0.051	0.068	0.085	0.099
ONPG series + IPTG	E	0.5	0.1375	0.458	0.02	0.035	0.047	0.06	0.069
ONPG series + IPTG	F	0.25	0.06875	0.458	0.011	0.019	0.025	0.031	0.036

Table S3. Initial velocities and reciprocals for the Lineweaver–Burk analysis, related to Figure 4
Reference dataset, Part III. V_0 in ΔA_{415} min^{-1} and in µmol ONP min^{-1} per assay (× 3.447); 1/[S] in mM^{-1}; $1/V_0$ in min $µmol^{-1}$. †, origin-assumed 0–2 min average rate (A_{415} = 0 assumed at t = 0); other values, regression on the initial linear window.

Series	Tube	[S] (mM)	2 min	4 min	6 min	8 min	10 min	V_0 (ΔA_{415}/min)	V_0 (µmol/min)	1/[S]	$1/V_0$
no inhibitor	A	2.2	0.282	0.546	0.815	1.094	1.328	0.132	0.455	0.45	2.2
no inhibitor	B	1.1	0.202	0.383	0.516	0.628	0.723	0.101†	0.348†	0.91	2.87
no inhibitor	C	0.55	0.156	0.278	0.357	0.425	0.468	0.078†	0.269†	1.82	3.72
no inhibitor	D	0.275	0.11	0.172	0.211	0.24	0.301	0.055†	0.190†	3.64	5.28
no inhibitor	E	0.1375	0.116	0.148	0.166	0.171	0.18	0.058†	0.200†	7.27	5
no inhibitor	F	0.06875	0.047	0.067	0.077	0.084	0.09	0.0235†	0.081†	14.55	12.35
+ 0.458 mM IPTG	A	2.2	0.198	0.48	0.586	0.784	0.966	0.099†	0.341†	0.45	2.93
+ 0.458 mM IPTG	B	1.1	0.065	0.127	0.196	0.239	0.294	0.03275	0.113	0.91	8.86
+ 0.458 mM IPTG	C	0.55	0.045	0.086	0.128	0.154	0.187	0.02075	0.072	1.82	13.98
+ 0.458 mM IPTG	D	0.275	0.029	0.051	0.068	0.085	0.099	0.0087	0.030	3.64	33.35
+ 0.458 mM IPTG	E	0.1375	0.02	0.035	0.047	0.06	0.069	0.00615	0.021	7.27	47.18
+ 0.458 mM IPTG	F	0.06875	0.011	0.019	0.025	0.031	0.036	0.0031	0.011	14.55	93.59

Table S4. Substrate converted by 2 and 10 min, related to Discussion
Percent of nominal ONPG converted to ONP, from $A_{415}/(\varepsilon \times 2/3)$ relative to the nominal reaction concentration, for every curve of both datasets.

Dataset	Tube	[ONPG] (mM)	A_{415} 2 min	% converted, 2 min	A_{415} 10 min	% converted, 10 min
Reference Part III (+IPTG)	A	2.2	0.198	2.6	0.966	12.6
Reference Part III (+IPTG)	B	1.1	0.065	1.7	0.294	7.7
Reference Part III (+IPTG)	C	0.55	0.045	2.4	0.187	9.8
Reference Part III (+IPTG)	D	0.275	0.029	3	0.099	10.3
Reference Part III (+IPTG)	E	0.138	0.02	4.2	0.069	14.4
Reference Part III (+IPTG)	F	0.0687	0.011	4.6	0.036	15
Reference Part III (-IPTG)	A	2.2	0.282	3.7	1.33	17.3
Reference Part III (-IPTG)	B	1.1	0.202	5.3	0.723	18.9
Reference Part III (-IPTG)	C	0.55	0.156	8.1	0.468	24.4
Reference Part III (-IPTG)	D	0.275	0.11	11.5	0.301	31.4
Reference Part III (-IPTG)	E	0.138	0.116	24.2	0.18	37.6
Reference Part III (-IPTG)	F	0.0687	0.047	19.6	0.09	37.6
Experimental Part III (+IPTG)	A	2.2	0.163	2.1	0.78	10.2
Experimental Part III (+IPTG)	B	1.1	0.269	7	1.11	28.9
Experimental Part III (+IPTG)	C	0.55	0.08	4.2	0.362	18.9
Experimental Part III (+IPTG)	D	0.275	0.041	4.3	0.188	19.6
Experimental Part III (-IPTG)	A	2.2	0.259	3.4	1.06	13.9
Experimental Part III (-IPTG)	B	1.1	0.224	5.8	1.06	27.8
Experimental Part III (-IPTG)	C	0.55	0.2	10.4	0.88	46
Experimental Part III (-IPTG)	D	0.275	0.202	21.1	0.764	79.8
Experimental Part III (-IPTG)	E	0.138	0.111	23.2	0.346	72.3

Table S5. Kinetic parameters under five initial-velocity estimators, related to Figure S3
Reference dataset. A, all five points of every curve; R, initial-window regression for every curve; S, origin-assumed 0–2 min average rate for every curve; O, the two-tier rule under the baseline-offset scenario; P, the two-tier rule of the Results. Negative values mark non-physical estimates from a negative Lineweaver–Burk intercept; –, undefined (nonlinear +IPTG fit ran to the K_m bound).

Estimator	Method	K_m	$K_{m,app}$	V_{max}	$V_{max,app}$	$K_{m,app}/K_m$	$V_{max,app}/V_{max}$	K_i	R^2 (−I)	R^2 (+I)	Physical	Primary
A	Lineweaver-Burk	3.405	1.317	0.849	0.214	0.39	0.25	-0.747	0.965	0.988	TRUE / TRUE	–
A	Nonlinear MM	20.9	–	4.744	–	–	–	–	0.995	0.951	TRUE / FALSE	–
O	Lineweaver-Burk	0.551	1.726	0.577	0.277	3.13	0.48	0.215	0.917	0.988	TRUE / TRUE	–
O	Nonlinear MM	0.524	–	0.527	–	–	–	–	0.924	0.971	TRUE / FALSE	–
P	Lineweaver-Burk	0.314	1.757	0.477	0.282	5.6	0.59	0.1	0.928	0.988	TRUE / TRUE	yes
P	Nonlinear MM	0.363	–	0.494	–	–	–	–	0.919	0.962	TRUE / FALSE	–
R	Lineweaver-Burk	-3.48	1.722	-0.912	0.276	-0.49	-0.3	-0.307	0.99	0.989	FALSE / TRUE	–
R	Nonlinear MM	3.379	–	1.147	–	–	–	–	0.998	0.972	TRUE / FALSE	–
S	Lineweaver-Burk	0.322	0.62	0.487	0.188	1.93	0.39	0.495	0.928	0.984	TRUE / TRUE	–
S	Nonlinear MM	0.436	–	0.537	–	–	–	–	0.908	0.954	TRUE / FALSE	–
P	Global nonlinear competitive	0.435	2.297	0.529	0.529	5.28	1	0.107	0.919	–	TRUE / NA	–

Table S6. Global fits of three inhibition models to both series, related to Figure 4
Twelve single measurements (six per series). Competitive and pure noncompetitive models have three parameters each; the mixed model has four and nests both.

Model	Parameter	Estimate	SE	RSS	AIC	n	Converged	Note
global competitive	Vmax	0.5285	0.072	0.01941	-35.07	12	TRUE	–
global competitive	Km	0.4351	0.1662	0.01941	-35.07	12	TRUE	–
global competitive	Ki	0.1071	0.0418	0.01941	-35.07	12	TRUE	–
global noncompetitive	Vmax	0.6262	0.1388	0.03713	-27.28	12	TRUE	–
global noncompetitive	Km	0.7438	0.3824	0.03713	-27.28	12	TRUE	–
global noncompetitive	Ki	0.4333	0.1806	0.03713	-27.28	12	TRUE	–
global mixed	Vmax	0.5286	0.0778	0.01942	-33.06	12	TRUE	–
global mixed	Km	0.4352	0.1788	0.01942	-33.06	12	TRUE	–
global mixed	Ki	0.1072	0.0792	0.01942	-33.06	12	TRUE	–
global mixed	Kip	500	422500	0.01942	-33.06	12	TRUE	at upper bound (model reduces to competitive)

Table S7. Kinetics from our own substrate series, related to Figure S4
Same rule and units as the Results. With the anomalous inhibited tube B excluded, the nonlinear fits show the pattern expected for competitive inhibition (apparent V_{max} ratio 1.01, a single-measurement estimate); including it makes the Lineweaver–Burk analysis meaningless.

Inhibited B	Method	K_m (mM)	$K_{m,app}$ (mM)	V_{max}	$V_{max,app}$	$V_{max,app}/V_{max}$	K_i (mM)	R^2 (−I)	R^2 (+I)	Tubes (−I / +I)
excluded ABCDE / ACD	Lineweaver-Burk	0.156	3.087	0.421	0.792	1.88	0.024	–	0.95	0.969
excluded ABCDE / ACD	Nonlinear MM	0.137	1.206	0.409	0.414	1.01	0.059	0.029	0.942	0.984
included ABCDE / ABCD	Lineweaver-Burk	0.156	38.16	0.421	9.042	21.5	0.002	–	0.95	0.945
included ABCDE / ABCD	Nonlinear MM	0.137	0.783	0.409	0.458	1.12	0.097	0.215	0.942	0.541
excluded ABCDE / ACD	Global nonlinear competitive	0.137	1.179	0.409	0.409	1	0.061	0.013	0.981	–
included ABCDE / ABCD	Global nonlinear competitive	0.149	0.647	0.419	0.419	1	0.137	0.115	0.691	–